Magnetic field lines

1. Magnetic field lines are continuous closed curves. The direction of magnetic field lines is from North pole to South pole outside the magnet and from South pole to North pole inside the magnet.
2. The direction of magnetic field at any point on the curve is known by drawing tangent to the magnetic field lines at that point.
3. Magnetic field lines never intersect each other. Otherwise, the magnetic compass needle would point towards two different directions, which is not possible.
4. The degree of closeness of the field lines determines the relative strength of the magnetic field. The magnetic field is strong where magnetic field lines crowd and weak where magnetic field lines are well separated.

(d) Magnetic flux

The number of magnetic field lines crossing any area normally is defined as magnetic flux Φ_B through the area. Mathematically, the magnetic flux through a surface of area $\vec{A}$ in a uniform magnetic field $\vec{B}$ is defined as

$$\Phi_B = \vec{B}.\vec{A} = BA\cos\theta = B_\perp A \qquad (3.6)$$

where θ is the angle between $\vec{B}$ and $\vec{A}$ as shown in Figure 3.8.

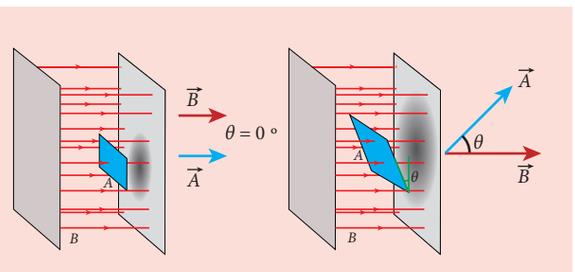


Figure 3.8 Magnetic flux

Special cases

(a) When $\vec{B}$ is normal to the surface i.e., $\theta = 0°$, the magnetic flux is $\Phi_B = BA$ (maximum).

(b) When $\vec{B}$ is parallel to the surface i.e., $\theta = 90°$, the magnetic flux is $\Phi_B = 0$.

Suppose the magnetic field is not uniform over the surface, the equation (3.6) can be written as

$$\Phi_B = \int \vec{B}.\mathrm{d}\vec{A}$$

Magnetic flux is a scalar quantity. The SI unit for magnetic flux is weber, which is denoted by symbol Wb. Dimensional formula for magnetic flux is $\left[ML^2T^{-2}A^{-1}\right]$. The CGS unit of magnetic flux is maxwell.

1 weber = 10^8 maxwell

The magnetic flux density is defined as the number of magnetic field lines crossing per unit area kept normal to the direction of lines of force. Its unit is Wb m^{-2} or tesla (T).

(e) Uniform magnetic field and Non-uniform magnetic field

Uniform magnetic field

Magnetic field is said to be uniform if it has same magnitude and direction at all the

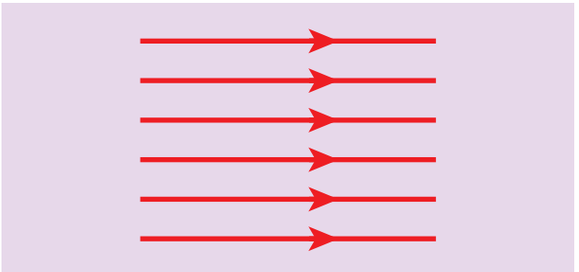


Figure 3.9 Uniform magnetic field

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